Very predictable Exam One: Nov 5th in class.

Tanks, linear first-order ODEs, separable ODEs, and first-order constant-coefficient linear systems. All including y_c, y_p, and initial conditions.

- Four questions on two sheets totaling 50 points.
- Questions like HW with fewer boxes stuck together.
- Write anything you want on a two-sided notecard less than 4×6 inches.
- Almost all review this week.

Separable and Linear ODEs

Separable $y' = g(t)h(y)$

- Multiply by dt and separate variables $\frac{dy}{h(y)} = g(t)\,dt$
- Integrate, then use the initial condition to determine C.
- Do not solve for y on an exam.

Linear $a_1(t)y' + a_0(t)y = f(t) \implies y' + p(t)y = q(t)$

- Divide by $a_1(t)$ to get the standard form $y' + p(t)y = q(t)$
- Multiply by the integrating factor $\mu(t) = \exp\left(\int p(t)\,dt\right)$ to get $(\mu y)' = \mu q(t)$.
- Integrate then use the initial condition to determine C.

Linear Systems $y' = Ay + f$: $y = y_c + y_p$ for $A \in \mathbb{R}^{n \times n}$

Complementary solution $y_c' = Ay_c$

- Find n linearly independent real solutions $\boldsymbol{y}_k(t)$ from eigenvalues and eigenvectors.
- Real eigenpairs give $e^{\lambda t}\boldsymbol{v}$. Splitting complex eigenpairs gives two LI solutions. Then $\boldsymbol{y}_c = \sum_{k=1}^{n} c_k \boldsymbol{y}_k$.

Particular solution $y_p' = Ay_p + f(t)$

- Substitute the undetermined coefficient form matching $\boldsymbol{f}$ and match coefficients.
- Add particular solutions for separate forcing terms.

For $\boldsymbol{y}(0) = \boldsymbol{y}_0$, determine $\boldsymbol{c}$ by solving $\boldsymbol{Mc} = \boldsymbol{y}_0 - \boldsymbol{y}_p(0)$, where $\boldsymbol{M} = [\boldsymbol{y}_1(0) \ \cdots \ \boldsymbol{y}_n(0)]$.

Exam Structure

In some order

- A tank problem
- A linear first-order ODE
- A separable first-order ODE
- A first-order constant-coefficient system
- A first-order constant-coefficient system

No eigenvalue computation required. No linear solves required. Required integration techniques include parts and substitution.

Variation of Parameters for $y' + p(t)y = q(t)$ with $y_1' = -p(t)y_1$ and $y_c(t) = cy_1(t)$

- **Variation of Parameters** replaces the constant c by a function $u(t)$.
- Substitute $y_p(t) = u(t)y_1(t)$ and of course $y_p' = u'y_1 + uy_1'$ into the ODE

$$y_p' + p(t)y_p = u'y_1 + u\underbrace{\left(y_1' + p(t)y_1\right)}_{0} = q(t)$$

So $u'(t)y_1(t) = q(t) \quad \Longrightarrow u(t) = \int \frac{q(t)}{y_1(t)}\,dt + C$

- The general solution is $y(t) = u(t)y_1(t) = \left(\int \frac{q(t)}{y_1(t)}\,dt + C\right) y_1(t)$.

$$\boxed{y(t) = \left(C + \int \frac{q(t)}{y_1(t)}\,dt\right) y_1(t)}$$

Variation of Parameters for $y' = Ay + f(t)$ with $\Phi' = A\Phi$ and $y_c = \Phi c$

- The **Fundamental Matrix** has n LI complementary solutions as columns

$$\Phi(t) = [\boldsymbol{y}_1(t) \ \cdots \ \boldsymbol{y}_n(t)]$$

so $\Phi' = [\boldsymbol{y}_1' \ \cdots \ \boldsymbol{y}_n'] = [A\boldsymbol{y}_1 \ \cdots \ A\boldsymbol{y}_n] = A[\boldsymbol{y}_1 \ \cdots \ \boldsymbol{y}_n] = A\Phi$

- Replace the constants in $\boldsymbol{y}_c = \Phi \boldsymbol{c}$ by $\boldsymbol{u}(t)$. Substitute $\boldsymbol{y} = \Phi \boldsymbol{u}$ into the ODE

$$\Phi' \boldsymbol{u} + \Phi \boldsymbol{u}' = A\Phi \boldsymbol{u} + \boldsymbol{f}$$

$$\underbrace{(\Phi' - A\Phi)}_{0} \boldsymbol{u} + \Phi \boldsymbol{u}' = \boldsymbol{f}$$

- So $\boldsymbol{u}' = \Phi^{-1} \boldsymbol{f}$ and $\boldsymbol{u}(t) = \boldsymbol{c} + \int \Phi(t)^{-1} \boldsymbol{f}(t)\, dt$.

$$\boxed{\boldsymbol{y}(t) = \Phi(t) \left(\boldsymbol{c} + \int \Phi(t)^{-1} \boldsymbol{f}(t)\, dt \right)}$$